

Sahil Wankhade

AI/ML Engineer

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EDUCATION

Rajiv Gandhi Institute of Technology

Mumbai, India

B.E. in Information Technology

Aug 2023 – May 2027 (Expected)

Relevant Coursework: Artificial Intelligence & Machine Learning, Data Structures & Algorithms, Database Management Systems, Operating Systems, Computer Networks, Software Engineering

TECHNICAL SKILLS

AI/ML: Retrieval-Augmented Generation (RAG), Cross-Encoder Reranking, LLM Evaluation, Benchmarking, Ablation Studies, PyTorch, scikit-learn, NumPy

Languages: Python, TypeScript, JavaScript, SQL

Web / Full-Stack: ReactJS, Next.js, Node.js, FastAPI, React Native, Tailwind CSS

Tools & Platforms: Git, GitHub, GitLab CI/CD, Hugging Face, Jupyter, Maestro (E2E), Firebase, VSCode

PROJECTS

Autonomous Research Agent (Python, Claude API, Tavily, Pydantic)

GitHub — Blog

- Built an autonomous research agent that decomposes complex queries, retrieves web evidence, ranks sources for credibility, extracts cited claims, and synthesizes a final report.
- Designed a structured multi-stage pipeline with schema-validated outputs, making each stage independently testable and debuggable.
- Evaluated 5 variants (2 baselines, 2 ablations, full system), finding iterative deepening improved its correct-claim rate from 75% to 96%.

Repo Assistant (Python, FastAPI, Claude API, Chroma)

GitHub — Live Demo

- Built and deployed an open-source RAG web app that answers natural-language questions about any public GitHub repo, citing exact files and line ranges.
- Built a retrieval pipeline (semantic embeddings + cross-encoder reranking over a Chroma vector store) with streaming ingestion and token-by-token answers.
- Built a recall@k evaluation harness (80% recall@3) to validate retrieval, and added prompt-injection defenses treating all retrieved code as untrusted input.

WORK EXPERIENCE

Junior Full Stack Intern, Bajaao

Mumbai, India

July 2025 – Current

- Built and shipped a cross-platform retail audit app (iOS, Android, Web) that replaced a paper-based store auditing workflow.
- Authored Maestro end-to-end test suites covering critical user flows and wired them into GitLab CI with Android-emulator provisioning and JUnit reporting.
- Collaborated with the team using Git, GitLab, and GitHub for version control, code reviews, and feature integration.

WRITING

“Building a Research Agent Taught Me Why Ablations Matter” — Medium. Technical article describing the design and evaluation of a research agent, including benchmark construction, ablation studies, and a negative experimental result.